



Convergence and Divergence Exhibit Different Response Characteristics to Symmetric Stimuli

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The dynamic characteristics of horizontal convergence and divergence eye movement responses to symmetric stimuli were examined. Binocular eye movements were recorded in five, visually normal adult subjects using the infrared reflection technique for symmetric convergent and divergent blur-free, disparity-only, step stimuli of 2, 4, 8, 12, and 16 deg. The main sequence as well as other temporal parameters including latency, time-to-peak velocity, time constant, and total duration were analyzed. A number of fundamental differences in the response characteristics were found between convergence and divergence. First, the slope of the peak velocity vs amplitude curve was approximately twice as high for convergence than divergence. The results are consistent with neurophysiological findings in monkeys and most findings in humans. Second, the initial fast component for convergence exhibited a larger amplitude than for divergence. This may reflect differences in central neural gain for convergence and divergence. And, third, all temporally related components were shorter for convergence than divergence. These findings provide an overall framework for vergence control and suggest fundamental differences in neural processing delays and neural controller pathways for convergence and divergence. © 1997 Elsevier Science Ltd. All rights reserved.

Convergence Divergence Main sequence Motoneuronal control Neurophysiology of eye movements

INTRODUCTION

Vergence is a disjunctive rotational movement of the two eyes such that the points of reference on the globes move in opposite directions (Schapiro *et al.*, 1968). Previously, we investigated the dynamic characteristics of symmetric horizontal convergence eye movements under a variety of stimulus conditions both in free-space and instrument-space (Hung *et al.*, 1994). It was found that the initial fast component of all movements fell along the normal main sequence for convergence steps, thus indicating a commonality in basic vergence dynamics under a variety of conditions. Furthermore, this result suggested that the neural controller program was dependent only on the response amplitude.

In our review of previous work on human vergence dynamics, it was found that with two exceptions (Erkelens *et al.*, 1989; Collewyn *et al.*, 1995), the slope§ was

higher for convergence than divergence (Rashbass & Westheimer, 1961; Bahill *et al.*, 1975a,b; Bahill & Stark, 1979; Bahill, 1981; Tenjin, 1981; Hung *et al.*, 1986, 1994; Zee *et al.*, 1992). Moreover, results in monkeys were consistent with these findings (Gamlin & Mays, 1992; Maxwell & King, 1992).

These differences among investigators may have been due to variations in experimental protocol or target configuration. For example, the range of disparity stimuli used by Erkelens *et al.* (1989) was ± 35 deg (starting at ± 5 deg), whereas that of the other investigators was not greater than ± 10 deg. Also, Erkelens *et al.* (1989) used physical targets in space whereas most of the other investigators projected a target on a screen that was shifted passively. Although Hung *et al.* (1994) also used physical targets in space as well as simultaneously presented targets on a screen, they only reported on convergence responses.

In contrast to the main sequence slope results, in which there was near unanimous agreement among the various researchers, there were a variety of differences between our study and some earlier ones. Previous research on directional differences in response amplitude between convergence and divergence did not separate vergence into fast and slow components (Hung *et al.*, 1994; Semmlow *et al.*, 1994). Instead, the amplitudes were based on the *final* overall or total (i.e. fast plus slow components) response level, with the small steady-state

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§When the slope of the vergence peak velocity vs amplitude relationship, or main sequence, was not given by the authors, it was estimated from the published data. The slope ranged from 3.3 to 4.0 (deg/sec)/deg for convergence and 1.3 to 2.1 (deg/sec)/deg for divergence.

variations being constrained by the limits (± 5 min arc) of Panum's fusional area at the fovea (Mitchell, 1966). Thus, no dynamic analysis of convergence vs divergence initial component amplitudes has been conducted. In addition, previous studies on the difference in response latency between convergence and divergence exhibited a dichotomy. Semmlow and Wetzel (1979) found a shorter latency for convergence (180 msec) than divergence (200 msec), with relatively small variability among subjects. On the other hand, Krishnan *et al.* (1973) found a longer latency for convergence (250 msec) than divergence (210 msec), with the latter showing a greater degree of variability. The discrepancy between these two studies may be due to the method for determining the beginning of the response. Semmlow and Wetzel (1979) used a subjective threshold criterion, whereas Krishnan *et al.* (1973) used a projection method in which a straight line was drawn from the "constant velocity portion" of the response position trace to the time axis. However, it may be difficult to determine precisely this portion of a continuously changing exponential-like response. Also, convergence and divergence trajectories have rather different dynamic shapes. Hence, such a method may not have provided an accurate measure of latency, especially for such small latency differences. Thus, latency differences remain an open question.

Other parameters, such as time constant and total duration of the overall response, were found to be shorter for convergence (195 and 750 msec, respectively) than divergence (240 and 1000 msec, respectively) (Zuber & Stark, 1968; Krishnan *et al.*, 1977; Semmlow & Wetzel, 1979). These parameters were included in this study for completeness and for comparison with previous results. The time constant has been used historically by some investigators to provide a quantitative, first-order estimate of the dynamics of the average vergence response to a step of disparity (Zuber & Stark, 1968; Semmlow & Wetzel, 1979). A faster movement will have a shorter time constant. For an exponential curve fitted to the overall shape of the response, the time constant is that time required for the response to reach 63% of its final steady-state level. Moreover, Bahill and Stark (1979) pointed out that there was a dependence of convergence response duration on stimulus amplitude, especially for amplitudes >10 deg, with duration being longer for the larger stimulus step amplitudes.

Thus, in the present paper, the detailed dynamic response characteristics for both convergence and divergence were investigated in humans. The results clearly demonstrated that the main sequence slope was consistently higher for convergence than divergence by a factor of *ca* 2, and further that the initial step response amplitude was larger for convergence than divergence. In addition, the other temporal parameters were all shorter for convergence than divergence. Finally, our findings provide a basis for resolving the conflicting results in many of the previous studies.

METHODS

Apparatus

The apparatus for stimulating vergence eye movements consisted of an optical system called the dynamic binocular stimulator (DBS). The details of this optical arrangement have been provided elsewhere (Semmlow & Venkiteswaran, 1976; Hung *et al.*, 1994). Basically, vertical luminous line targets (1 deg high and 0.25 deg wide) were presented on an oscilloscope screen (phosphor type P31) positioned 28.5 cm from a point P in the DBS. Point P was optically conjugate to the entrance pupils of the subject's eyes. A variable aperture was located at P, so that reducing its diameter to 0.5 mm provided blur-free viewing (Ward & Charman, 1987). The single image of a target T was divided in the DBS optical pathway and seen by both eyes, thus providing the percept of a single vertical line in depth. Any change in lateral position of the target on the oscilloscope screen resulted in equal and opposite target displacements in the two eyes, thus providing pure symmetric disparity stimulation in the horizontal dimension only. Stimulus generation, eye movement recording, and data analysis were controlled by a Hewlett-Packard 9000, Series 550 computer. Horizontal eye movements were recorded binocularly using a Skalar infrared eye movement monitor (Model 6500). This eye movement system had a linear range of ± 25 deg, a resolution of 5 min arc, and a bandwidth of 200 Hz. Data were analyzed off-line and graphed on a Laserjet plotter.

Procedure

Five subjects participated in the experiments. Their ages ranged from 28 to 52 yr. Two of the subjects (GH and JS) had extensive experience in eye movement experimentation, whereas the other three subjects had minimal experience. The research followed the tenets of the Declaration of Helsinki. Informed consent was obtained after the nature and possible consequences of the study were explained to the subjects, and the research was approved by the Rutgers University Institutional Review Board.

For each subject, symmetric vergence eye movements were recorded for convergence (+) and divergence (−) disparity-only step stimuli of 2, 4, 8, 12, and 16 deg. The range of the stimulus was limited to ± 16 deg due to the limits of target displacement on the oscilloscope screen. For each disparity step stimulus, 10–15 trials were recorded. The baseline reference level was adjusted prior to the experimental trials by rotating the two mirrors in front of the eyes (Hung *et al.*, 1994). It was set for 8 deg of convergence in the trials that contained mostly convergent stimuli, and 16 deg of convergence in the trials that contained the larger divergent stimuli. This was done because some subjects found it difficult to fuse targets that were more divergent than 0 deg (i.e. beyond optical infinity), which required a relative divergence response (Borish, 1970). A three-point calibration was performed following each response, and it consisted of steps of equi-angular target displacement over the

stimulus range. When the subject remained in the bite-bar apparatus, there was no evidence of drift greater than the 5 min arc resolution between a previous- and present-trial calibration over the 2.5 sec response duration. During an experimental session, which consisted of *ca* 50 trials lasting 1 hr, both stimulus direction and amplitude were randomized. Each subject participated in either two or three sessions.

Analysis of data

In the off-line data analysis procedure, the calibrated movements of each eye were subtracted to yield the overall vergence response. Records containing artifacts such as blinks or saccades during the initial portion of the dynamic response phases were deleted. Only those responses that remained within the range used for calibration (± 16 deg) were retained for further analysis.

For the velocity calculation, a zero-phase finite impulse response (FIR) filter was used (Usui & Amidror, 1982; Behrens & Weiss, 1992). The digital filter was a three-point central difference algorithm with values taken at $\pm \Delta T$, $\pm 2\Delta T$, $\pm 3\Delta T$, with $\Delta T = 5$ msec. Each difference value was multiplied by a weight, and the three weighted differences were summed to give the velocity at time T . The weights were calculated to give a digital filter frequency response which was equal to the ideal differentiator, with no additional phase shift. Based on preliminary studies, the threshold criteria* for determining the beginning and end of a vergence response were velocities greater or less than 4 deg/sec, respectively. Cursors were then used to mark these points, as well as the peak velocity location. This was then verified by visual inspection and accepted or modified manually as needed.

Other parameters were also determined to assess quantitatively differences between convergence and divergence. These included latency, time to peak velocity, time constant, and total duration, which were all measured relative to the response onset. The time constant was obtained as the best fit, as determined by the least squared error, between the first-order model and the response at a resolution of 5 msec, for the overall dynamic portion of the response [i.e. combined fast plus slow components (Hung *et al.*, 1994)]. Such a measure was used for consistency with earlier estimates of the overall time constant of the averaged response (Zuber & Stark, 1968; Semmlow & Wetzel, 1979).

The method introduced and used in a previous paper

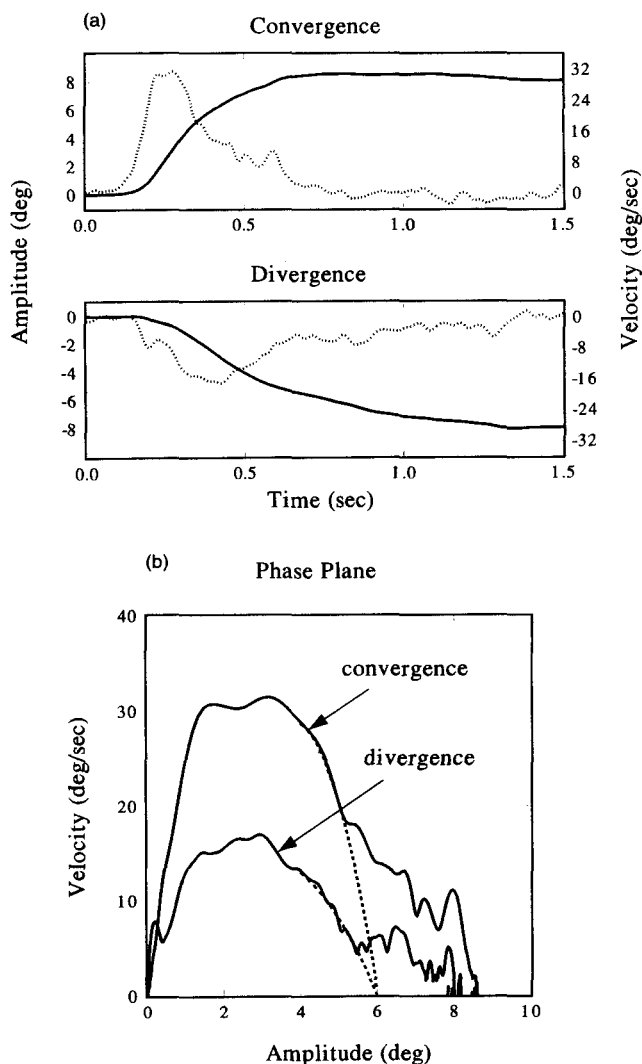


FIGURE 1. (a) Amplitude (solid line) and velocity (dashed line) time courses for representative convergence (+) and divergence (−) responses having approximately the same fast component amplitude. (b) Phase-plane plots for the responses in (a) showing absolute value of instantaneous velocity vs. amplitude for convergence and divergence. The phase-plane is useful for determining where changes in dynamics occur. Extrapolated (dashed line) fast component amplitude was used in the data analysis. Note that although these convergence and divergence movements have the same fast component amplitude (*ca* 6 deg), the convergence peak velocity is nearly twice that found for divergence. Subject JD.

(Hung *et al.*, 1994) for the assessment of vergence dynamics was used in the present study. It is based on the fact that vergence responses contain two components: an initial larger and faster open-loop pre-programmed component, which is presumably directly related to the midbrain neuronal controller signal, followed by a smaller and slower closed-loop visual feedback-based component, which reduces the residual disparity to within neurosensory tolerances, i.e. Panum's fusional area (Panum, 1858; Hung *et al.*, 1986; Semmlow *et al.*, 1986, 1994). To ascertain the actual amplitude of the fast component, which could be masked by the subsequent slow component, the vergence movement was displayed as a computer-generated phase-plane (instantaneous

*Based on the 200 Hz sampling rate, the instantaneous velocity over one sampling interval has a lower limit of 16.7 deg/sec. However, the post-recording computer velocity algorithm estimates the velocity over six sampling intervals, i.e. $\pm 3\Delta T$ about the measurement point. Thus, a slow velocity of $5 \text{ min arc}/(6 \times 5 \text{ msec}) = 2.8 \text{ deg/sec}$ could be detected. Moreover, it must be pointed out that once the movement begins, the velocity rises quickly, and thus the error in the detection of the beginning of the movement is quite small. In addition, both the position and the velocity traces are also visually inspected to verify the beginning and end of the movement.

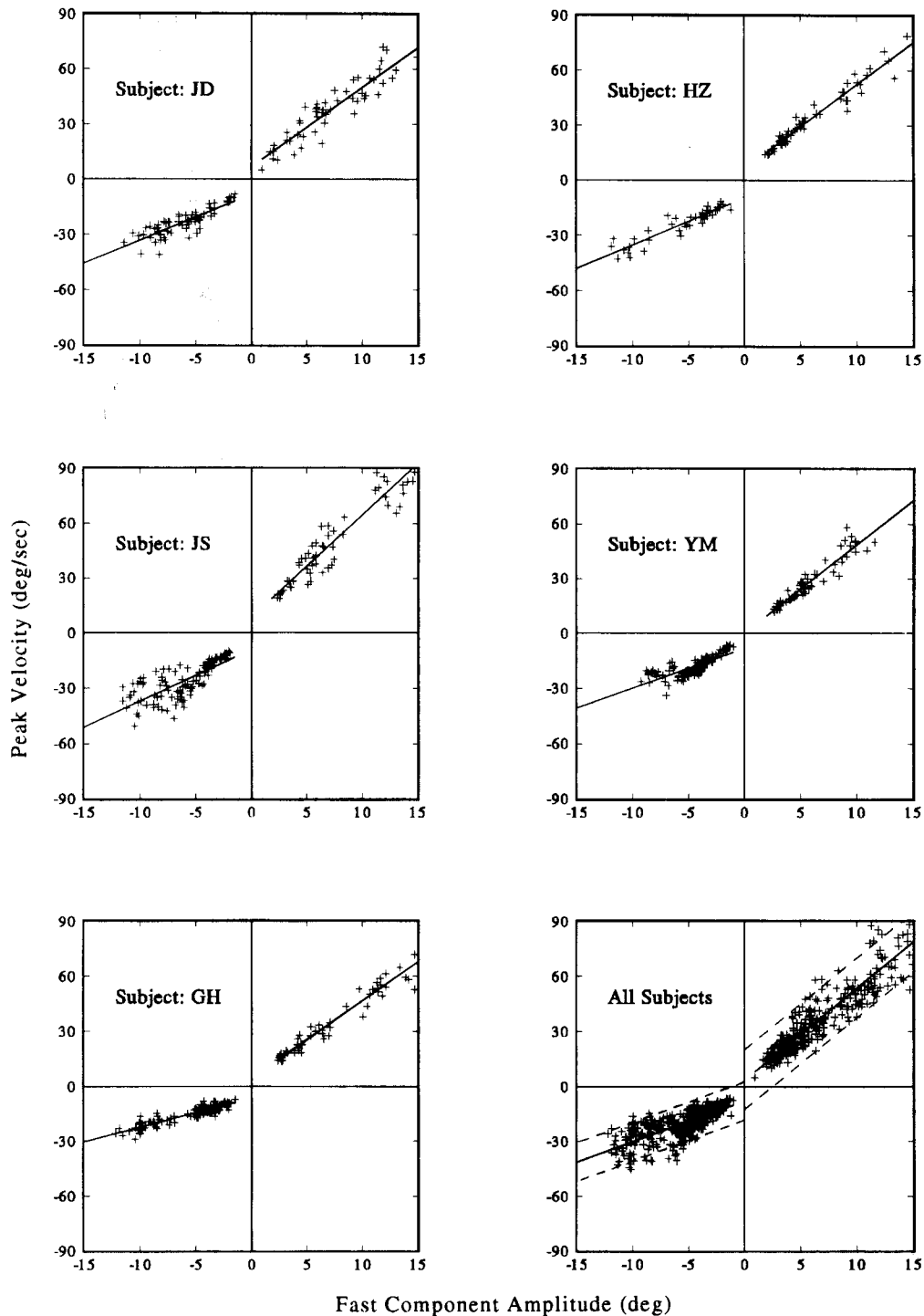


FIGURE 2. Composite plot of main sequence, or peak velocity vs amplitude data points, for symmetric vergence step responses for each of the five subjects and all subjects combined (lower right subplot) showing higher slope of linear regression line (solid line) for convergence (positive amplitudes and peak velocities) than for divergence (negative amplitudes and peak velocities). The 95% confidence interval is shown as dashed lines in the subplot for all subjects combined.

velocity vs position) plot (Banks, 1986). Examples of typical convergence and divergence eye movement amplitude and velocity traces are shown in Fig. 1(a). Their corresponding phase-planes are shown in Fig. 1(b). For the phase-plane trace, the region from just after the peak velocity to just before the break to the slow component portion was subjectively delimited manually by means of cursors. The computer program used a

second-order polynomial algorithm to fit a least squares curve to the phase-plane trace within this region. The computer-generated curve was then extrapolated to the intersection at the position axis [see dashed line in Fig. 1(b)]. The magnitude at the intersection was considered to represent the amplitude of the initial fast component for that response. The peak velocity was also obtained from the same phase-plane plot. The amplitude and peak

velocity values were obtained for each vergence movement, and then graphed on a main sequence plot (Bahill *et al.*, 1975a,b; Bahill & Stark, 1979; Bahill, 1981). Also, the initial fast component amplitude was plotted against the stimulus step amplitude to determine the relative contribution of the initial component to the overall step amplitude for both convergence and divergence.

RESULTS

The paired peak velocity vs amplitude values, or main sequence, for convergence and divergence responses of all subjects combined is shown in the lower right subplot of Fig. 2. The group mean least square regression lines are also shown, along with their 95% confidence intervals. The linear regression equation* was $y = 4.99x + 3.73$ deg/sec for convergence and $y = 2.24x - 7.65$ deg/sec for divergence, with r values of 0.92 and 0.75 for convergence and divergence, respectively. The ratio of mean convergence-to-divergence slope was 2.23. Thus, the mean convergence slope was more than twice that found for divergence, and this difference was statistically significant ($t = 10.8$, d.f. = 4, $P < 0.0002$).

The individual main sequence data and least square regression lines for convergence and divergence for the five subjects are also shown in Fig. 2. The ratios of individual subject convergence-to-divergence slopes ranged from 1.73 to 2.62. The r values ranged from 0.92 to 0.97 for convergence and from 0.76 to 0.93 for divergence. For each of the r values, the correlation was significant (d.f. ≥ 90 , $P < 0.01$). Also, the relatively high r^2 values (0.85–0.94 for convergence and 0.57–0.86 for divergence) demonstrated that much of the variability can be accounted for by the relation between the two main sequence parameters. There was no evident effect of age or experience on the ratio of the slopes among the subjects.

The initial component amplitude is plotted as a function of stimulus step amplitude for both convergence and divergence in Fig. 3. The linear regression equations for the mean values were $y = 0.71x + 1.27$ deg and $y = 0.52x + 1.00$ deg for convergence and divergence, respectively. It can be seen from the figure that the initial fast component amplitude for convergence was *ca* 25% larger than that for divergence for all stimulus amplitudes, and this difference was statistically significant ($t = 3.21$, d.f. = 4, $P < 0.003$). Also, for stimulus amplitudes >4 deg, the initial amplitudes for convergence and

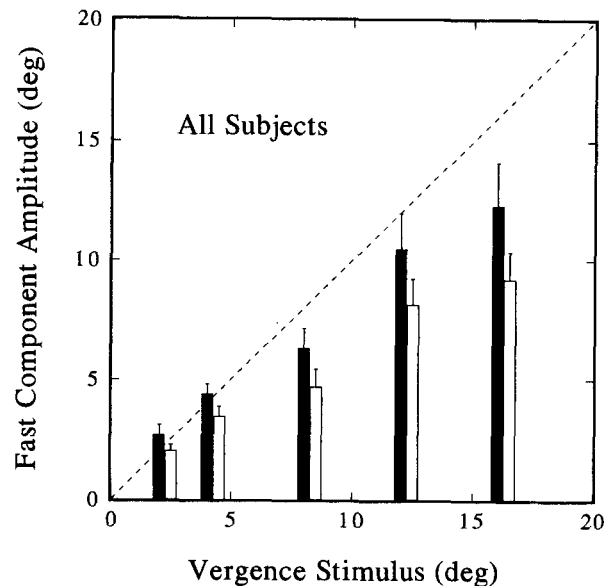


FIGURE 3. Initial fast component amplitude vs stimulus disparity step amplitude. The linear regression equations for the mean values are $y = 0.71x + 1.27$ deg and $y = 0.52x + 1.00$ deg, for convergence (filled bars) and divergence (open bars), respectively. Bars represent the mean value for data on all subjects, and error bars represent ± 1 SEM.

divergence were *ca* 80% and 60%, respectively, of the stimulus amplitude.

There were also statistically significant differences between mean convergence and divergence responses across subjects for the four temporal parameters. They were all shorter for convergence than for divergence (see Fig. 4 and Table 1). In addition, there was a dependence on the stimulus amplitude for both the time constant [convergence: $F(4,20) = 10.1$, $P < 0.0001$; divergence: $F(4,20) = 14.6$, $P < 0.0001$] and total duration [convergence: $F(4,20) = 7.8$, $P < 0.0006$; divergence: $F(4,20) = 13.3$, $P < 0.0001$], with their values increasing with increased stimulus amplitude. In contrast, latency [convergence: $F(4,20) = 0.7$, $P > 0.6$; divergence: $F(4,20) = 0.3$, $P > 0.9$] and time to peak velocity [convergence: $F(4,20) = 2.2$, $P > 0.1$; divergence: $F(4,20) = 1.3$, $P > 0.3$] did not show a dependence on stimulus amplitude (see Fig. 4 and Table 1).

DISCUSSION

Differences between convergence and divergence

There were six significant differences between convergence and divergence found in the present study. First, the slope was higher for convergence than divergence by a factor of about 2. A ratio being significantly greater than one is consistent with the estimated ratios of 1.57 for Rashbass and Westheimer (1961) and 1.75 for Zee *et al.* (1992). The results are also consistent with recent neurophysiological findings in the midbrain of the monkey by Gamlin and Mays (1992) and Mays and colleagues (Mays, 1984; Mays & Porter, 1984; Mays *et al.*, 1986). For example, examination of the medial rectus motoneuronal signal time courses presented by Gamlin

*The intercept is used only as a parameter in the linear regression equation and is not to be interpreted as the peak velocity at zero amplitude. Although small clamped open-loop step disparities (Rashbass & Westheimer, 1961) can elicit a response by initially triggering the slow component (Hung *et al.*, 1986), small closed-loop step disparities, i.e., amplitude < 0.5 deg, do not elicit fast component step responses (Semmlow *et al.*, 1986). The existence of this fast component threshold indicates that there is no closed-loop step response within this range. In some of the sub-plots, the minimum response amplitude that is elicited is somewhat above this threshold range. See Fig. 2.

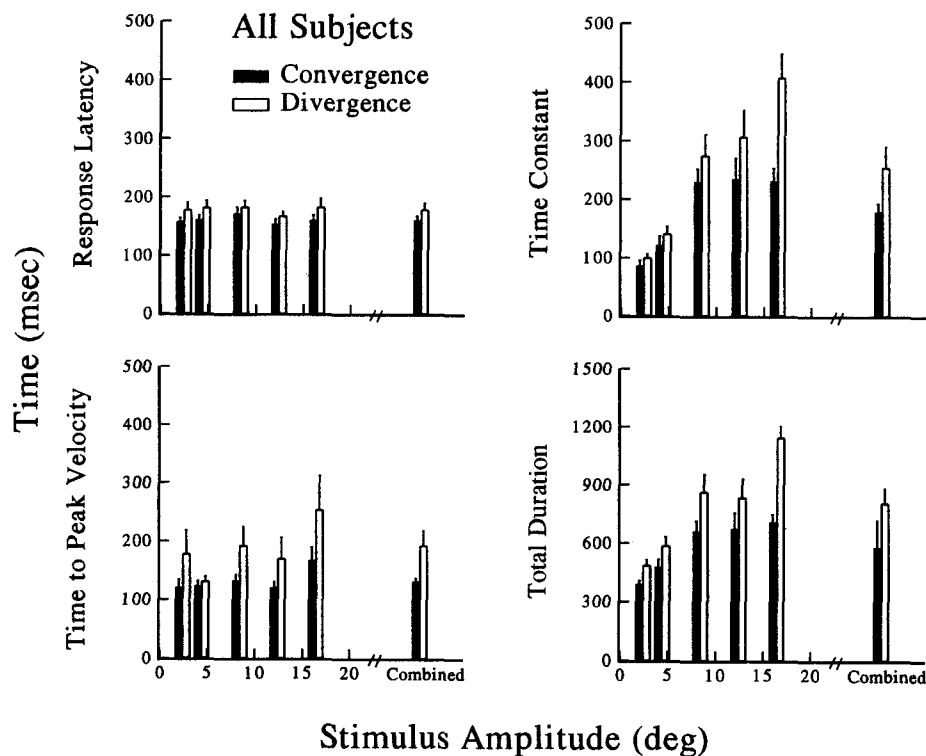


FIGURE 4. Composite plot of four temporal parameters: latency, time-to-peak velocity, time constant, and total duration as a function of stimulus step amplitude (2, 4, 8, 12, 16, and combined of all the stimuli) for convergence (filled bars) and divergence (open bars). Values (in msec) are mean + 1 SEM.

TABLE 1. Temporal parameter values (mean $\pm$ 1 SEM, in msec) for convergence and divergence for all subjects (*t*-test for difference between convergence and divergence; one-way ANOVA for amplitude dependence)

		Stimulus step amplitude (deg)						Amplitude dependence
		2	4	8	12	16	Combined	
Latency	Conv.	158 $\pm$ 6	161 $\pm$ 7	172 $\pm$ 10	154 $\pm$ 8	161 $\pm$ 8	161 $\pm$ 7	NS
	Div.	177 $\pm$ 13	181 $\pm$ 12	182 $\pm$ 12	167 $\pm$ 8	182 $\pm$ 16	178 $\pm$ 12	NS
	Diff.	-19	-20	-10	-13	-21	-17	
	<i>P</i>	≤ 0.05	≤ 0.05	≤ 0.05	< 0.05	NS	≤ 0.05	
Time to peak velocity	Conv.	120 $\pm$ 13	123 $\pm$ 8	132 $\pm$ 10	120 $\pm$ 10	167 $\pm$ 22	131 $\pm$ 5	NS
	Div.	177 $\pm$ 41	130 $\pm$ 9	191 $\pm$ 33	169 $\pm$ 37	253 $\pm$ 59	192 $\pm$ 26	NS
	Diff.	-57	-7	-59	-49	-86	-61	
	<i>P</i>	NS	NS	≤ 0.05	NS	NS	≤ 0.05	
Time constant	Conv.	86 $\pm$ 9	121 $\pm$ 16	230 $\pm$ 22	235 $\pm$ 35	231 $\pm$ 22	179 $\pm$ 14	$P \leq 0.05$
	Div.	99 $\pm$ 7	140 $\pm$ 13	274 $\pm$ 37	306 $\pm$ 46	407 $\pm$ 41	254 $\pm$ 37	$P \leq 0.05$
	Diff.	-13	-19	-44	-71	-176	-75	
	<i>P</i>	NS	NS	NS	NS	≤ 0.05	≤ 0.05	
Total duration	Conv.	389 $\pm$ 17	478 $\pm$ 37	662 $\pm$ 52	674 $\pm$ 82	709 $\pm$ 39	580 $\pm$ 138	$P \leq 0.05$
	Div.	484 $\pm$ 30	584 $\pm$ 48	860 $\pm$ 93	832 $\pm$ 96	1141 $\pm$ 62	804 $\pm$ 78	$P \leq 0.05$
	Diff.	-95	-106	-198	-158	-432	-224	
	<i>P</i>	≤ 0.05	≤ 0.05	≤ 0.05	NS	≤ 0.05	≤ 0.05	

Note: For simplicity, *P* values ≤ 0.05 are designated by $P \leq 0.05$ and those > 0.05 are designated by NS.

and Mays (1992) revealed that the slope was higher for convergence than divergence. The motoneuronal signal time course was also shown to be associated with the velocity of the vergence movement. This signal is integrated to provide the step portion of the controller signal (Mays *et al.*, 1986). The neurophysiological results along

with our present findings suggest that the pulse component amplitude is considerably smaller for the slower divergence movements. Such a difference in neural controller programs has recently been found in our laboratory using a new physiologically based computer simulation model of the human vergence system, with

divergence exhibiting essentially a step-only controller signal.

Second, the initial fast component amplitude was consistently larger (*ca.* 25%) than that found for convergence than divergence. This may reflect a difference in the central neural computation of the required motor signal, or gain, for the initial step amplitude, with divergence underestimating the amplitude considerably more than convergence in the initial phase of its control strategy. The initial component gains for convergence and divergence were *ca.* 0.8 and 0.6, respectively. Underestimation of the error input appears to be a general neural control strategy for the oculomotor system. For example, with the saccadic system, hypometria is the norm and hypermetria is the exception, with hypometria apparently being a purposeful behavior to optimize computation time (Baloh & Hornubia, 1976). The logic of such control is that any secondary error will then only require an amplitude computation, and not both amplitude and direction, thereby saving time in gaining final target acquisition. The underestimation of target error seen in the saccadic system also appears to be true for disparity vergence, with its preprogrammed initial fast component typically being hypometric and its slow component correction being in the same direction under direct visual feedback control. The amplitude gain asymmetry, moreover, points to a dynamic directional asymmetry in disparity vergence control, which is not unexpected (see later clinical discussion on static asymmetries).

Third, latency was shorter for convergence than divergence. This is consistent with the results of Semmlow and Wetzell (1979). It suggests different neural processing pathways and/or processing components for convergence and divergence.

Fourth, the time-to-peak velocity was shorter for convergence than divergence. That is, the peak velocity was attained earlier in the response for convergence than for divergence. This may be related to differences in motoneuronal controller signals, with convergence having a pulse-step signal and divergence exhibiting essentially a step signal (Mays, 1984; Mays & Porter, 1984; Mays *et al.*, 1986).

Fifth, the time constant was shorter for convergence than divergence. This is consistent with previous findings (Zuber & Stark, 1968; Krishnan *et al.*, 1977; Semmlow & Wetzell, 1979). It reflects differences between convergence and divergence both in the relative contribution of the fast component signal to the overall response as well as in the slow component fusional process.

Sixth and finally, the total duration of the overall response was shorter for convergence than divergence. The difference in the time-to-peak velocity between convergence and divergence was 61 msec (see Table 1), which approximated the difference in their fast component durations. On the other hand, the difference in total duration was 224 msec (see Table 1). This indicates that much of the difference in the total duration of responses reflects the difference in the slow component dynamics

between convergence and divergence ($224 - 61 = 163$ msec).

Control process for the initial component

Latency and time-to-peak velocity were independent of stimulus amplitude (see Fig. 4 and Table 1). This was consistent with the notion that the calculation of controller signal magnitude required the same amount of time regardless of the amplitude. Independence of time-to-peak velocity suggested that the initial fast open-loop component could be approximately represented by a linear process.* Indeed, a decade ago, Hung *et al.* (1986) modeled the open-loop component as a second-order linear process, resulting in realistic simulations of experimental responses under a variety of stimulus conditions. On the other hand, both the time constant and total duration of the *overall* response were dependent on stimulus amplitude. These parameters were interrelated: a response with a larger time constant would have a longer overall response. Such a dependence on amplitude reflected the effects of: (1) the dynamic interactions between the initial fast component and the subsequent slow component, where the initial open-loop component provided a larger proportion of the movement amplitude and the later closed-loop component provided the remainder of the response; and (2) the slow component fusional process, which tended to be slower for larger amplitude movements. This is consistent with the suggestion by Bahill and Stark (1979) relating the dependence of convergence duration on stimulus amplitude.

Consistency with clinical findings

The directional response asymmetries found in the present paper are consistent with several clinical findings (Schor & Ciuffreda, 1983; Ciuffreda, 1992). First, relative fusional vergence ranges are larger for convergence than for divergence (Borish, 1970; Hung & Ciuffreda, 1994). Second, relative fusional convergence ranges are more easily expanded with oculomotor training both in visually normals (Daum, 1986a,b; Ciuffreda & Tannen, 1995) and symptomatic abnormals (Grisham, 1983) than are relative fusional divergence ranges. And, third, the phoria/fixation disparity relation is considerably steeper for convergence than divergence (Ogle, 1950).

Measurement of responses to symmetric horizontal disparities

In our experimental apparatus, we were able to present accurate horizontal symmetric step-disparity stimuli to

*For a linear process, the shape of the dynamic time course remains the same independent of the amplitude (D'Azzo & Houpis, 1988). This is applicable to the initial fast component time course, which is closely related to the time to peak velocity. Indeed, this parameter does not show a dependence on stimulus amplitude (see their Table 1). The time constant, however, encompasses both the fast and slow components, and it does show a dependence on stimulus amplitude. Hence the overall time course does not exhibit linear system characteristics.

the subjects. No vertical disparity was introduced in the stimulus configuration, and hence vertical eye movements were not recorded. The horizontal vergence responses of the subjects were generally smooth and exhibited relatively few saccades during the initial fast component portion of the movement. In the present data analysis, the few movements that exhibited saccades or other artifacts such as blinks were deleted. Such relatively saccade-free responses are consistent with the reports by Rashbass and Westheimer (1961), Zuber and Stark (1968), Semmlow and Wetzel (1979), and Littmann (1989). For example, Littmann (1989) presented vergence responses to symmetric step disparity stimuli in seven subjects. He noted that in six of the seven subjects it was possible (p. 50) "... to observe the perfect motion symmetry of the two eyes, a symmetry which is also maintained at low contrast". Very few saccades are evident in the numerous movement responses he presented. Littmann noted for the one subject who did not exhibit such symmetric smooth responses that, "We can immediately note the zig-zags which the left eye makes toward the periphery shortly after the start of the movement. This asymmetry is caused by a strong dominance of the subject's right eye."

Feedback and stability

Regarding the role of visual feedback in the initial portion of the response, this should be considered in terms of the relatively long response time as compared to its latency. Our combined group results indicated that the latency was 161 and 178 msec for convergence and divergence, respectively, and that the step response total duration was considerably longer (three to four times) than the latency. If visual feedback were to play a role in determining the size of the initial step, this would lead to severe difficulties in maintaining stability, as the instantaneous error during the fast component movement cannot be acted upon until one latency interval later, when the actual instantaneous error would be substantially different from the intended movement for correcting the earlier detected error. For example, let us consider an 8 deg disparity step stimulus and the response at *ca* 0.3 sec after stimulus onset [see Fig. 1(a)]. The response amplitude is 4 deg, and hence the error is 4 deg. If a correction for this error is made, it would take *ca* 160 msec before the corrective response is initiated. However, by this time, the ongoing vergence movement would be about 7 deg, resulting in a 1 deg residual error. The 4 deg convergence movement that was initiated at 0.3 sec would now cause an overshoot. This would lead to dynamic response instability. One possible way to maintain stability and still use visual feedback is to have a very low controller gain (Hung *et al.*, 1986). However, this would result in a much smaller and slower overall response than found experimentally (Semmlow *et al.*, 1986). On the other hand, the dual-mode model for vergence (Hung *et al.*, 1986; Semmlow *et al.*, 1994) uses a fast open-loop initial component to attain the initial fast dynamics. After completion of the initial movement,

when the disparity, or vergence error, is small, the slow closed-loop component takes over; since it has much lower gain and slower dynamics, it can maintain stability in the presence of visual feedback.

Functional significance

Lastly, in terms of possible functional significance, convergence may be more important than divergence, as it directs the eyes toward nearby objects. These nearby objects can then be manipulated by the hands for gathering, feeding, or detailed viewing. In addition, because manipulation of near objects is useful, it would be advantageous to bring them towards oneself more quickly. On the other hand, far objects have a less immediate impact, especially for survival purposes, and therefore are relatively less important at any given moment. Thus, there is less need to direct the eyes towards them as rapidly. Hence, the faster convergence dynamics may be a reflection of their greater functional significance in daily life.

REFERENCES

- Bahill, A. T. (1981). *Bioengineering: Biomedical, medical, and clinical engineering* (pp. 154–157). Englewood Cliffs, NJ: Prentice–Hall.
- Bahill, A. T., Clark, M. R. & Stark, L. (1975a) Computer simulation of overshoot in saccadic eye movements. *Computer Programs in Biomedicine*, 4, 230–236.
- Bahill, A. T., Clark, M. R. & Stark, L. (1975b) Glissades-eye movements generated by mismatched components of the saccadic motoneuronal control signal. *Mathematical Biosciences*, 26, 303–318.
- Bahill, A. T. & Stark, L. (1979). The trajectory of saccadic eye movements. *Scientific American*, 240, 108–114.
- Baloh, R. W. & Hornubia, V. (1976). Reaction time and accuracy of the saccadic eye movements of normal subjects in a moving-target task. *Aviation Space & Environmental Medicine*, 47, 1165–1167.
- Banks, S. (1986). *Control systems engineering* (pp. 263–267). Englewood Cliffs, NJ: Prentice-Hall.
- Behrens, F. & Weiss, L. R. (1992). An algorithm separating saccadic from nonsaccadic eye movements automatically by use of the acceleration signal. *Vision Research*, 32, 889–893.
- Borish, I. M. (1970). *Clinical refraction*, 3rd. ed. Chicago, IL: Professional Press.
- Ciuffreda, K. J. (1992). Components of clinical near vergence testing. *Journal of Behavioral Optometry*, 3, 3–13.
- Ciuffreda, K. J. & Tannen, B. (1995). *Eye movement basics for the clinician*. St. Louis, MO: Mosby Year-Book.
- Collewijn, H., Erkelens, C. J. & Steinman, R. M. (1995). Voluntary binocular gaze-shifts in the plane of regard: Dynamics of version and vergence. *Vision Research*, 35, 3335–3358.
- Daum, K. M. (1986a) Double-blind placebo-controlled examination of timing effects in the training of positive vergences. *American Journal of Optometry and Physiological Optics*, 63, 807–812.
- Daum, K. M. (1986b) Negative vergence training in humans. *American Journal of Optometry and Physiological Optics*, 63, 487–496.
- D'Azzo, J. J. & Houpis, C. H. (1988). *Linear control system analysis and design, conventional and modern* (pp. 87–91). New York: McGraw-Hill.
- Erkelens, C. J., Van der Steen, J., Steinman, R. M. & Collewijn, H. (1989). Ocular vergence under natural conditions. II. Gaze shifts between real targets differing in distance and direction. *Proceedings of the Royal Society, London (B)*, 236, 441–465.
- Gamlin, P. D. R. & Mays, L. E. (1992). Dynamic properties of medial rectus motoneurons during vergence eye movements. *Journal of Neurophysiology*, 67, 64–74.

- Grisham, J. D. (1983). Treatment of binocular dysfunction. In Schor, C. M. & Ciuffreda, K. J. (Eds), *Vergence eye movements: Basic and clinical aspects* (pp. 605–646). Boston: Butterworths.
- Hung, G. K. & Ciuffreda, K. J. (1994). Sensitivity analysis of relative accommodation and vergence. *IEEE Transactions on Biomedical Engineering*, 41, 241–248.
- Hung, G. K., Ciuffreda, K. J., Semmlow, J. L. & Horng, J.-L. (1994). Vergence eye movements under natural viewing conditions. *Investigative Ophthalmology and Visual Science*, 35, 3486–3492.
- Hung, G. K., Semmlow, J. L. & Ciuffreda, K. J. (1986). A dual-mode dynamic model of the vergence eye movement system. *IEEE Transactions on Biomedical Engineering*, 33, 1021–1028.
- Krishnan, V. V., Farazian, F. & Stark, L. (1973). An analysis of latencies and prediction in the fusional vergence system. *American Journal of Optometry and Archives of the American Academy of Optometry*, 50, 933–939.
- Krishnan, V. V., Shirachi, D. & Stark, L. (1977). Dynamic measures of vergence accommodation. *American Journal of Optometrics and Physiological Optics*, 54, 470–473.
- Littmann, E. (1989). Vergence and contrast. *Optometrie*, 3, 43–58.
- Maxwell, J. S. & King, W. M. (1992). Dynamics and efficacy of saccade-facilitated vergence eye movements in monkeys. *Journal of Neurophysiology*, 68, 1248–1260.
- Mays, L. E. (1984). Neural control of vergence eye movements: Convergence and divergence neurons in the midbrain. *Journal of Neurophysiology*, 51, 1091–1108.
- Mays, L. E. & Porter, J. D. (1984). Neural control of vergence eye movements: Activity of abducens and oculomotor neurons. *Journal of Neurophysiology*, 52, 743–761.
- Mays, L. E., Porter, J. D., Gamlin, P. D. R. & Tello, C. A. (1986). Neural control of vergence eye movements: Neurons encoding vergence velocity. *Journal of Neurophysiology*, 56, 1007–1021.
- Mitchell, D. E. (1966). A review of the concept of "Panum's Fusional Areas". *American Journal of Optometry and Archives of the American Academy of Optometry*, 43, 387–401.
- Ogle, K. N. (1950). *Researches in binocular vision* (pp. 59–93). New York: Hafner.
- Panum, P. L. (1858). *Physiologische Untersuchungen über das Sehen mit zwei Augen*. Kiel: Schwesche Buchhandlung.
- Rashbass, C. & Westheimer, G. (1961). Disjunctive eye movements. *Journal of Physiology (London)*, 159, 339–360.
- Schapiro, M., Cline, D. & Hofstetter, H. W. (Eds) (1968). *Dictionary of visual science*. Philadelphia, PA: Chilton Book Co.
- Schor, C. M. & Ciuffreda, K. J. (Eds) (1983). *Vergence eye movements: Basic and clinical aspects*. Boston: Butterworths.
- Semmlow, J. L., Hung, G. K. & Ciuffreda, K. J. (1986). Quantitative assessment of disparity vergence components. *Investigative Ophthalmology and Visual Science*, 27, 558–564.
- Semmlow, J. L., Hung, G. K., Horng, J.-L. & Ciuffreda, K. J. (1994). Disparity vergence eye movements exhibit preprogrammed motor control. *Vision Research*, 34, 1335–1343.
- Semmlow, J. L. & Venkiteswaran, N. (1976). Dynamic accommodative vergence in binocular vision. *Visual Research*, 16, 403–411.
- Semmlow, J. L. & Wetzel, P. (1979). Dynamic contributions of the components of binocular vergence. *Journal of the Optical Society of America*, 69, 639–645.
- Tenjin, S. (1981). Study on fusional convergence. *Folio Ophthalmologica Japonica*, 32, 800–808.
- Usui, S. & Amidror, I. (1982). Digital low-pass differentiation for biological signal processing. *IEEE Transactions on Biomedical Engineering*, 29, 686–693.
- Ward, P. A. & Charman, W. N. (1987). On the use of small artificial pupils to open-loop the accommodation system. *Ophthalmology and Physiological Optics*, 7, 191–193.
- Zee, D. S., Fitzgibbon, E. J. & Optican, L. M. (1992). Saccade-vergence interactions in humans. *Journal of Neurophysiology*, 68, 1624–1641.
- Zuber, B. L. & Stark, L. (1968). Dynamical characteristics of the fusional vergence eye-movement system. *IEEE Transactions on Systems Science and Cybernetics*, 4, 72–79.

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